Section 4: first mathematical review

Proposed passages for author review

November 11, 2026

Status. Historical supporting document: the review destination is now the working manuscript, where the proposed passages appear in dark orange awaiting author review. Source: repository checkpoint `82fe1a1`, original annotated PDF pp. 36–43. Main-text statements were read independently before the annotations. The accompanying Lean file checks the algebraic and ordered-set calculations below; it does not yet prove the numbered retractions or their concrete Grothendieck constructions.

1. Cell dimensions and homogeneous notation (R056–R059)

Remark 4.2, proposed replacement for the cube description. Choose a finite cell decomposition $X = \coprod_j C_j$ and set

$$\chi_g(X) = \sum_j (-1)^{\dim C_j}.$$

Equivalently, a rectilinearization uses open boxes of varying dimensions, including dimension zero. A singleton already requires the zero-dimensional case. Independence of decomposition, invariance and multiplicativity remain proof obligations; this correction is not a substitute for those proofs.

Notation 4.3, proposed clarification. In the graded ring write $u = [\{1\}]_1$ and $L_1 = [\mathbb{G}_m]_1 + u$; the ring unit is $1 = [\{1\}]_0$. Graded localizations by $[\mathbb{A}]$ mean localization by L_1. The ideal $!I[*]$ is generated by the homogeneous degree-one differences $[\gamma^\sharp]_1 - [\mathbb{G}_m]_1$. In the ungraded ring the corresponding affine-line class is $L = [\mathbb{G}_m] + 1$.

Display (4.2), proposed replacement. With $T = [\mathbb{G}_m]_1$, the calculation must read

$$\mathbb{E}_g([\mathrm{RV}^{\circ\circ}] + 1 - u) = -TL_1^{-1} + 1 - uL_1^{-1} = 0.$$

Indeed $L_1 = T + u$. The printed last term $-L_1^{-1}$ omits the degree-one point factor; its algebraic remainder is $(u-1)L_1^{-1}$. After forgetting degree, the point is 1 and the familiar ungraded expression is recovered. The bounded version uses $1 - uu^{-1} = 0$. Component tensor maps are first bilinear over $\mathbb{Z}$; their sum must then respect the full homogeneous balancing relations (R058). This last step is a formalization obligation, not a change of target.

2. Degree-zero constants (R062)

Remark 4.6, proposed replacement. Let

$$D = \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}), \qquad \mathbb{Z}[D] \subseteq \mathrm{FN}(\Gamma, \mathbb{Z})$$

be the group ring spanned by the p_γ for all definable Gamma elements. Then $\mathrm{K}\mu\Gamma[0] \cong \mathbb{Z}[D]$, and the target of λ is $\mathbb{Z}[D] \oplus X\mathrm{FN}(\Gamma, \mathbb{Z})[X]$.

Reason. A degree-zero object is a finite collection of empty tuples with definable constant weights. A singleton of weight $\frac{1}{2}v(t)$ over $\mathbb{C}((t))$ is permitted even though this value is not in the parameter field's value group $\mathbb{Z}v(t)$. The earlier remark after the Gamma-category definition already distinguishes these two groups. This proposal makes the coefficient notation consistent with that remark. All three assignments for ψ keep their stated degree restrictions: only p_γ occurs in degree zero (R063).

3. The sign in Proposition 4.7 (R064)

Proposed third bullet.

$$\mu\mathbb{E}_g(\mathfrak{h}) = +[1]/([\mathbb{A}]), \qquad \mu\mathbb{E}_b(\mathfrak{h}) = 0.$$

Derivation from the paper's definitions. For $\boldsymbol{I} = ((0,\infty), -\mathrm{id})$, the corrected weight $\mu_I(x) = -x + x$ is zero. Thus

$$\lambda_g([\boldsymbol{I}]) = -p_0 X, \qquad \psi(-p_0 X) = -[\mathbb{G}_m]_1 = +[1]_1 \pmod{([\mathbb{A}])}.$$

The last equality is forced by $[\mathbb{A}]_1 = [\mathbb{G}_m]_1 + [1]_1$. For the bounded characteristic the open ray has characteristic zero. The original negative sign is not the value of this construction. This is a change to a proposition's stated value and needs author approval. It also requires checking later callers, including the use of 4.7 in §8.

4. Bounded support and negative intervals (R067–R068)

Notation 4.11, proposed replacement. Define $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ to consist of the functions in $\mathrm{FN}(\Gamma, \mathbb{Z})$ with doubly bounded nonzero locus. Set

$$o_\gamma = \begin{cases} \mathbf{1}_{(0,\gamma)}, & \gamma > 0, \\ \mathbf{1}_{(\gamma,0)}, & \gamma < 0, \\ -p_0, & \gamma = 0. \end{cases}$$

The existing two alternative convolution formulas are then read with explicit sign cases. In particular

$$o_\alpha * o_\beta = \begin{cases} -o_{\alpha+\beta}, & \alpha, \beta \geq 0 \text{ or } \alpha, \beta \leq 0, \\ -o_\alpha - o_\beta - p_0, & \alpha < 0 < \beta \text{ or } \beta < 0 < \alpha. \end{cases}$$

For $\beta \neq 0$, $p_\alpha * o_\beta$ is the characteristic function of the open interval with endpoints α and $\alpha + \beta$; for $\beta = 0$ it equals $-p_\alpha$.

Reason. Under the literal printed convention, $\gamma < 0$ gives the empty interval $(0, \gamma)$ and hence $o_\gamma = 0$. The proposed relation $o_\gamma + p_0 = 0$ would then kill p_0, the convolution unit, contradicting the claimed quotient $\mathbb{Z}$. The special convention $o_0 = -p_0$ concerns functions, not the actual empty interval $(0, 0)$.

5. Admissible radii and the zero generator (R069–R071)

Notation 4.13, proposed replacement. Define $\boldsymbol{P}_\gamma$ for nonnegative definable γ *admitting a definable* $t_\gamma \in \gamma^\sharp$. Explicitly, the index set is

$$\{\gamma \geq 0 : \ \exists t \in \mathrm{RV}(\mathrm{dcl}\,\mathbb{S}),\ \mathrm{vrv}(t) = \gamma\}.$$

Use this same index set for the generators of $(\boldsymbol{P}_\Gamma)$. This preserves the intended definition wherever its specified point exists. It does not assume a definable section of the value map.
Reason. The distinction is already explicit after the definition of Gamma-categories. Over $\mathbb{C}((t))$, the value $v(t)/2$ is definable, but its RV fibre has no definable point. For example, the conjugation sending $t^{1/2}$ to $-t^{1/2}$ changes its RV class. Such a radius cannot be used to select t_γ over the unchanged base.

Proposition 4.15, proposed proof replacement at the generator calculation. If $\gamma = 0$, the annulus is empty and $[\{t_0\}] - [1] = 0$ by the zero-volume, value-zero singleton isomorphism; thus $\boldsymbol{P}_0 = 0$. If $\gamma > 0$, its Gamma part is $(0, \gamma] = (0, \gamma) \amalg \{\gamma\}$ and

$$\chi((0,\gamma))[\mathbb{G}_m]_1 + \chi(\{\gamma\})[\mathbb{G}_m]_1 + [1]_1 - [1]_1 = -[\mathbb{G}_m]_1 + [\mathbb{G}_m]_1 = 0.$$

The displayed open-interval-plus-endpoint expansion in the original proof does not apply at zero; there it leaves an extra $[\mathbb{G}_m]_1$. This is a proof correction; the intended bounded retraction is retained.

6. Matters not approved or completed by this review

The affine-localized injectivity assertion in Remark 4.4 (R060) has a separate independent paper-level reconstruction in the companion file `R060-independent-review.md`, checked against the primary source. The generic localization mechanism is checked in Lean; the actual geometric and graded identifications are not yet formalized. The assertion has not been silently replaced by "natural homomorphism." The point-localized comparison, convolution associativity (R061), enlarged alternative quotient (R066), and diagram shorthand (R072) still need their actual formal constructions; they are not assumed false on account of annotations. Pure editorial changes R055, R065, R070 and R168 can be collected separately. No referee annotation has been marked resolved yet.

Author decision requested. Approve the proposed cell/homogeneity clarifications and mathematical corrections in items 1–5, or specify which passage should retain a different interpretation. The highest-impact decision is the sign in Proposition 4.7. Full coverage, remaining proof obligations and exact formalization scope are recorded in `section 4/COVERAGE.md`.